

Knowledge distillation

>> Section 1

>> Section 2

>> Section 3

>> Section 4

>> Section 5

>> Section 6

Research Notes on Knowledge distillation

Knowledge distillation

$$y_i(\mathbf{x} | \mathbf{t}) = e^{z_i(\mathbf{x})} / \sum_j e^{z_j(\mathbf{x})} \quad \{ \displaystyle y_i(\mathbf{x}) = \frac{e^{z_i(\mathbf{x})}}{\sum_j e^{z_j(\mathbf{x})}} \}$$

>> Section 7

Do until a desired level of sparsity or performance is reached: Train the network (by methods such as backpropagation) until a reasonable solution is obtained Compute the saliencies for each parameter Delete some lowest-saliency parameters Deleting a parameter means fixing the parameter to zero. The "saliency" of a parameter θ is defined as $\frac{1}{2} (\frac{\partial^2 L}{\partial \theta^2}) \theta^2$, where L is the loss function. The second-derivative $\frac{\partial^2 L}{\partial \theta^2}$ can be computed by second-order backpropagation. The idea for optimal brain damage is to approximate the loss function in a neighborhood of optimal parameter θ^* by Taylor expansion: $L(\theta) \approx L(\theta^*) + \frac{1}{2} \sum_i (\frac{\partial^2 L}{\partial \theta_i^2}(\theta^*)) (\theta_i - \theta_i^*)^2$ where $\nabla L(\theta^*) \approx 0$, since θ^* is optimal, and the cross-derivatives $\frac{\partial^2 L}{\partial \theta_i \partial \theta_j}$ are neglected to save compute. Thus, the saliency of a parameter approximates the increase in loss if that parameter is deleted.